Rafi Dahoui

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Education

Université de Montréal **Expected Graduation: April 2027**
Bachelor of Science (B.Sc.) in Computer Science *Montreal, QC*

Projects

LaunchGuard | *Source Code* **Java | Spring Boot | Kafka | React | PostgreSQL**

- Built an **incident-monitoring platform** that links system failures to recent deployments and presents findings through a React dashboard
- Used **Spring Boot, Kafka, and PostgreSQL** to process and store events, with OIDC authentication and role-based access control
- Implemented **187 automated tests** (144 Java, 43 frontend) using Testcontainers and Playwright, with GitHub Actions CI and Docker

Pothole Montréal – Mobile Reporting Platform **React Native | TypeScript | Supabase | PostGIS**

- Built a **mobile reporting app** allowing users to report potholes with photos, GPS locations, and interactive maps
- Implemented **PostGIS proximity queries**, duplicate detection, and reverse geocoding to improve reporting accuracy
- Developed a **Supabase backend** with Row-Level Security, private photo storage, and an admin moderation system

Autonomous Investment Research Lab **Python | Pytest | Backtesting | AI Agents**

- Built a **Python backtesting platform** to evaluate algorithmic trading strategies against SPY using historical data and virtual capital
- Implemented **walk-forward validation**, transaction costs, and risk controls to reduce look-ahead bias and evaluate strategy performance
- Developed an **AI research-agent pipeline** to generate and validate investment ideas, supported by **345 automated tests**

MaVille – City Works Management Platform **Java | MongoDB | Maven | JUnit**

- Developed the **Java/MongoDB backend** for user authentication, registration, and municipal service request management
- Integrated **Montréal roadwork and traffic APIs** and implemented backend unit tests using JUnit and Maven
- Collaborated with teammates to resolve frontend-backend integration issues and improve application functionality

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C, PHP
Frameworks & APIs: Spring Boot, React, React Native, Next.js, Node.js, REST APIs
Databases & Cloud: PostgreSQL, MongoDB, Supabase, PostGIS, Kafka, AWS, Docker
Testing & Tools: JUnit, Pytest, Playwright, Testcontainers, Git, GitHub Actions, Maven
Spoken Languages: English, French, Arabic

Work Experience

3 Amigos **Oct 2022 – Present**
Waiter *Montreal, QC*

- Managed customer requests across **10+ tables** while staying organized during high-volume shifts
- Worked closely with kitchen and bar teams to keep orders moving and resolve service issues during peak periods
- Communicated directly with customers to handle concerns and find quick solutions during busy shifts

Grillades Torino **Oct 2021 – Oct 2022**
Manager *Montreal, QC*

- Oversaw daily operations, coordinated staff, and handled customer and operational issues
- Managed cash handling and end-of-day reconciliation while maintaining payment accuracy
- Prioritized staff responsibilities and resolved issues during busy periods to keep service running smoothly